

A structured roadmap needed to achieve goals!



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Overall, it seems to be a good policy in terms of objectives but it seems to be quite silent on a structured roadmap to achieve these goals. The policy needs to be made more specific, and some more emphasis has to be laid on application engineering. This good initiative needs to be more comprehensive to suit wants of our country and also in line with socio-economic aspirations of our nation.

The draft policy mentions that attractive fiscal incentives across the value chain of the ESDM sector through a modified incentive package scheme will be offered. However, it is quite silent about the method of quantifying the same and also on the impact of changing government policies particularly in relation to the WTO requirements. For example, recently, DEPB Scheme has been withdrawn out of the WTO commitment. Before making any commitment to the industry on fiscal incentives, a detailed analysis and assurance has to be made to the industry that whatever we do is WTO-compliant.

In a sector like semiconductors where huge investments are required, the government or public-sector initiatives should become an intrinsic part of the strategy. Semiconductors have to be given the status of infrastructure or even more in order to attract investments in this high-cost sector. There should also be automatic approval of FDI without any limit and adequate protection should be provided to the investments being put here for an agreed-upon time.

Another important issue which needs more focus is R&D in a specific manner. Dedicated teams of experts from India or abroad have to be provided as a talent pool to all the semiconductor components manufacturers in a central outfit. The best talent from all over the world should be attracted and continuous upgradation of technology through proper research and application engineering should be the prime focus of this institution.

The policy mentions that requisite incentives may be provided by streamlining the procedures and logistics to facilitate imports of components, sub-systems and import of products. Such a policy framework has a high risk of being misutilised and at times works as a disincentive for investment into core technologies. Any facility provided for imports for the purpose of exports has to be tagged with some timeframe which should enable the local industry to develop such products both for domestic and export markets and also substantial value addition should happen in India which should be going up in a progressive manner from quarter to quarter.

For human resource development, the policy mentions creation of a specialised institute for semiconductor design. In my view, such specialised institutes should become part of the existing engineering institutions of repute and universities instead of being created from grassroot level, which is both time-consuming and costly. If a decision has to be taken for creating specialised institutes for semiconductor design, it should be with the R&D outfit which I have proposed earlier.

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including electronics manufacturing services industry

(b) DTA sales of ITA-1/zero-duty electronics products to be treated as physical exports and extending all the benefits of export schemes

(c) Create incentives for relocation to India of electronic hardware manufacturing units facing cost pressures in developed countries

(d) Globally market and showcase chip design, product design and embedded software industry capabilities

(e) Promote export of electronics products in emerging regions like Africa, South America and Asia among others

3. Human resource development

3.1 Work closely with the private sector, universities and other institutions of learning and design programmes to

ensure that adequate trained and skilled manpower is available to the industry

(a) Facilitate enhancement of the number of graduates and other skilled manpower by suitably increasing capacities in colleges/ITIs and polytechnics through public- and private-sector investments

(b) Support creation of capacities within academic institutions to enhance the production of adequate number of PhDs and postgraduates for supporting the growth of chip design and embedded software and board/hardware design industry in the country

(c) Encourage setting up of skill-oriented courses and training programmes for electronics designs along with hands-on laboratories enabling graduates from other disciplines to migrate to ESDM

(d) Creation of a specialised insti-

tute for semiconductor design

(e) Extend special manpower development programme for very large-scale integration (VLSI) chip design to include larger number of colleges and students leveraging the national knowledge network

(f) Create an institutional mechanism for the faculty development in various ESDM related subjects

(g) Collaborate with national and international institutions for development of new skills and courseware on latest manufacturing technologies and products in the ESDM sector

4. Developing and mandating standards

4.1 Curb inflow of sub-standard and unsafe electronics products by mandating technical and safety standards

(a) Develop Indian standards to meet specific Indian conditions including climatic, power supply and handling conditions by suitably reviewing existing standards

(b) Mandate technical standards in the interest of public health and safety

(c) Set up an institutional mechanism within Department of Information Technology for mandating compliance to standards for electronics products

(d) Develop a national policy framework for enforcement and use of standards and quality management processes

(e) Strengthen the lab infrastructure for testing of electronics products and encourage development of conformity-assessment infrastructure by private participation

(f) Create awareness amongst consumers against sub-standard and spurious electronics products

5. Cyber security

5.1 To create a complete secure cyber eco-system in the country, careful and due attention is required for creation of well-defined technology and systems, use of appropriate technology and more importantly development of appropriate products and solutions. The priorities for action will be suitable design and development of indigenous appropriate products through frontier technology-/product-oriented research, testing and validation of security of